

```
import random
```

```
import pandas as pd
```

```
g = 9.81
```

```
def calculate_theoretical_acceleration(m, M, mu_k):
```

```
    """
```

```
    Calculate theoretical acceleration using the formula:
```

```
    
$$a = (m * g - \mu_k * M * g) / (M + m)$$

```

```
    """
```

```
    return (m * g - mu_k * M * g) / (M + m)
```

```
def simulate_measured_acceleration(theoretical_acceleration):
```

```
    """
```

```
    Simulate measured acceleration by introducing a small random error.
```

```
    """
```

```
    error = random.uniform(-0.1, 0.1)
```

```
    return max(0, theoretical_acceleration + error)
```

```
def generate_observation_table(hanging_masses, cart_mass, mu_k, distance):
```

```
    """
```

```
    Generate an observation table for the experiment.
```

```
    Parameters:
```

```
        hanging_masses: List of hanging masses (kg).
```

```
        cart_mass: Mass of the cart (kg).
```

```
        mu_k: Coefficient of kinetic friction.
```

distance: Length of the bench (m).

Returns:

DataFrame with calculated results.

```
"""
```

```
results = []
```

```
for m in hanging_masses:
```

```
    theoretical_a = calculate_theoretical_acceleration(m, cart_mass, mu_k)
```

```
    measured_a = simulate_measured_acceleration(theoretical_a)
```

```
    theoretical_time = (2 * distance / theoretical_a) ** 0.5 if theoretical_a > 0  
else float('inf')
```

```
    measured_time = (2 * distance / measured_a) ** 0.5 if measured_a > 0  
else float('inf')
```

```
results.append({
```

```
    "Hanging Mass (kg)": m,
```

```
    "Cart Mass (kg)": cart_mass,
```

```
    "Coefficient of Friction": mu_k,
```

```
    "Distance (m)": distance,
```

```
    "Theoretical Acceleration (m/s^2)": theoretical_a,
```

```
    "Measured Acceleration (m/s^2)": measured_a,
```

```
    "Theoretical Time (s)": theoretical_time,
```

```
    "Measured Time (s)": measured_time,
```

```
})
```

```
return pd.DataFrame(results)
```

```
hanging_masses = [0.1, 0.2, 0.3, 0.4, 0.5]
cart_mass = 2.0
mu_k = 0.1
distance = 1.0
observation_table = generate_observation_table(hanging_masses, cart_mass,
mu_k, distance)
print(observation_table.to_string(index=False))
```

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